# EmberLink

## Linked fire alerts for rural lanes

Concept note | Seva First Innovation Challenge 2026 | Problem Statement 39 | Theme 5: Surakshit Bharat | Track A: Product and Technology

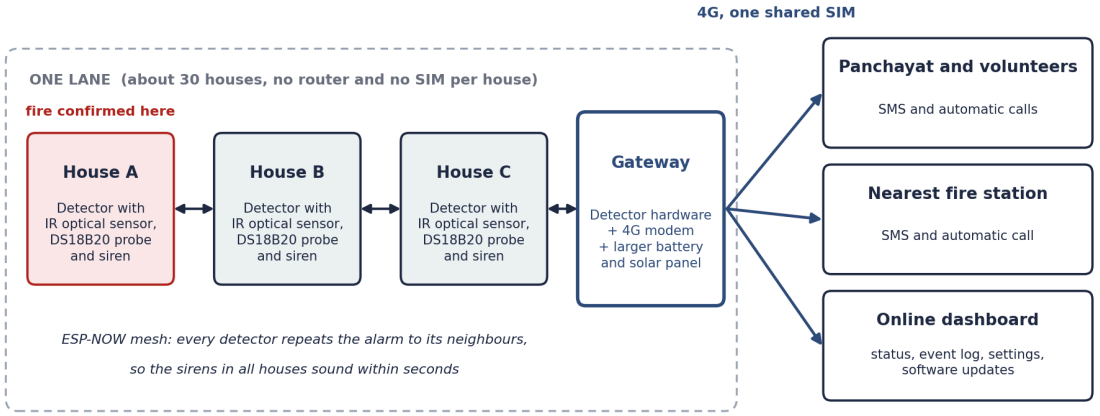
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## Problem and vulnerability profile

Houses in rural lanes are often built of thatch, bamboo, timber and mud, share walls or stand a few feet apart, and are lit and cooked in with open flame. The 2011 Census recorded that 15.0 per cent of Indian households lived under roofs of grass, thatch, bamboo, wood, mud or similar material. A fire that starts in one house can reach the next within minutes, usually at night, and it is noticed only when a neighbour smells smoke. A single-house smoke alarm wakes one family. A connected alarm with a modem and SIM per house costs more than a Panchayat can spend across a village. And the call for help still depends on a frightened person finding a phone.

## Proposed solution

EmberLink gives each house a low-cost solar-powered detector and gives the lane one link to the outside. Detectors use an optical infrared smoke sensor together with a DS18B20 temperature probe, which together are designed to ignore cooking smoke. A confirmed fire sounds the house siren and is passed from detector to detector over an ESP-NOW mesh, a direct radio link that needs no router or SIM, so every siren in the lane sounds within seconds. One unit, the gateway, adds a 4G modem. It sends SMS messages and places calls to the Panchayat, volunteers and the fire station, and reports to an online dashboard. The lane needs one SIM. The mesh, its automatic maintenance, remote software updates, remote device profiles and gateway selection already exist in an earlier implementation on ESP8266. This project ports it to ESP32 and adds the gateway bridge, dashboard, self-test, pilot and handover.




*System topology: detectors linked by the mesh, one gateway, one shared SIM.*

## Operational workflow

| 1. Detection | 2. Lane alert | 3. Dispatch |
|---|---|---|
| The optical channel and the temperature channel must agree before a lane alarm is raised. One channel alone gives a local pre-alarm and a log entry, which keeps cooking smoke from waking the lane. | The detecting unit sounds its siren and broadcasts an authenticated alert. Each neighbour repeats it once and drops duplicates. If a unit fails, the mesh routes around it. | The gateway sends SMS to every listed contact, then calls them in order until one replies OK. The event also goes to the dashboard. Authorised numbers can send STATUS, TEST, RESET and SILENCE. |

## Specification

| Subunit | Component | Purpose |
|---|---|---|
| **Processor and mesh radio** | ESP32-WROOM-32U | Dual-core, external antenna connector, ESP-NOW on the 2.4 GHz radio. No separate long-range radio. |
| **Smoke sensing** | Proprietary optical IR chamber | Existing design built to reject cooking smoke; dust-drift compensation in firmware. |
| **Heat sensing** | DS18B20 digital probe | Temperature channel that confirms or rejects the optical channel. |
| **Audible warning** | 12 V piezo siren with boost stage | Target at least 85 dB at 1 m, to wake sleeping residents. |
| **Detector power** | LiFePO4 18650 (2000 mAh), LiFePO4 solar charger, 5 V 2 W panel | Thermally stable chemistry; target average current at most 4 mA, which supports 14 days without sun. |
| **Gateway cellular** | SIM7670G 4G HAT, boost supply, 6000 mAh pack, 6 W panel | SMS, calls and data through one SIM; operator voice and SMS support verified in week 1. |
| **Enclosure** | Flame-retardant ABS, IP65 | Mounted on a non-combustible backing. |
| **Management software** | Web dashboard and cloud service | Lane map, event log, device profiles, staged software updates, roles for Panchayat, volunteer and engineer. |
| **Cost** | ₹1,450 per detector; ₹3,110 per gateway | A 30-house lane costs ₹45,160, or ₹1,505 per house. No household subscription. Planning estimates to be replaced by quotations. |

## Implementation and testing

| Stage | Dates | Work |
|---|---|---|
| **1. Field survey and baseline** | 25 Oct to 1 Nov | Choose and map a lane with the Gram Panchayat; record materials, spacing, signal strength; measure the existing build as a baseline. |
| **2. Port and build** | 25 Oct to 15 Nov | Port to ESP32; Rev B board; gateway bridge; dashboard; weekly self-test. |
| **3. Bench and campus lane** | 9 to 16 Nov | Ten-unit mock lane; timing, dispatch, self-healing, update and battery tests. |
| **4. Live lane pilot** | 17 to 27 Nov | Install gateway and ten detectors; tune thresholds against real cooking; drills; train the volunteer. |
| **5. Handover and exhibition** | 28 Nov to 15 Dec | Hand over to the Panchayat; live demonstration at the exhibition. |

## Targets we will measure in the pilot

| Measure | Target |
|---|---|
| False alarms | No more than 1 in 100 unit-days in homes that cook daily |
| Lane alert time | All sirens within 5 s of confirmation, about 1 s per hop at worst |
| Dispatch | SMS to every listed contact within 30 s in at least 95% of trials |
| Self-healing | Reports reach the gateway within 60 s after any one unit is switched off |
| Remote update | New software reaches every unit with no site visit; a failed update rolls back |
| Battery | Detector average current at most 4 mA; gateway at least 5 days without sun |

**Prior work and student contribution.** The mesh, update and profile machinery and the sensing method come from the mentor's earlier implementation. The students lead the gateway bridge, dashboard, self-test analytics, field survey, pilot, training and documentation, and share the ESP32 port and Rev B board with the mentor's engineers. EmberLink is an early-warning aid and does not replace the fire service.